DRIVEABILITY—ENGINE FLOODING, EXTENDED CRANK TIME, HARD STARTING, COLD ENGINE WITH AMBIENT TEMPERATURES 30-40 F (-1-5 C)	Article No. 02-16-1

FORD: 2002 THUNDERBIRD

LINCOLN: 2000-2002 LS

ISSUE

Some vehicles may exhibit an engine flooding, extended engine cranking time required to start, or hard start condition. These conditions would typically occur when the ambient temperatures are between 30-40° F (-1-5 °C), and humidity is greater than 75%. This may be caused by the PCM, PCM Calibration or the Idle Air Control (IAC) Valve.

ACTION

Verify the condition and service the vehicle using one of the following service procedures, depending on engine displacement and the PCM installed in the vehicle.

SERVICE PROCEDURE

2002 Model Year Vehicles - All Engines

2002 model year vehicles (all engines) require only a Powertrain Control Module (PCM) calibration update. If working on a 2002 model year vehicle, reprogram the PCM to the latest calibration level. Refer to the PCM Calibration Chart for details. No further service is required.

2001 Model Year Vehicles - V6 Engines

V6 2001 model year vehicles may require only a PCM reprogram or a new PCM, Idle Air Control (IAC) Jumper Harness, IAC, IAC Gasket and IAC Bolts depending on the PCM installed in the vehicle. Refer to “Obtaining the PCM Module Identification” section of this article for further details.

2001 Model Year Vehicles - V8 Engines

V8 2001 model year vehicles may require only a PCM reprogram or a new PCM and IAC Jumper Harness depending on the PCM installed in the vehicle. Refer to “Obtaining the PCM Module Identification” section of this article for further details.

2000 Model Year Vehicles - V6 Engines

V6 2000 model year vehicles require a new PCM, IAC Jumper Harness, IAC, IAC Gasket and IAC Bolts be installed in the vehicle. Refer to “Obtaining the PCM Module Identification” and “V6 Vehicles Requiring a New PCM” sections of this article for further details.

2000 Model Year Vehicles - V8 Engines

V8 2000 model year vehicles require a new PCM and IAC Jumper Harness be installed in the vehicle. Refer to “Obtaining the PCM Module Identification” and “V8 Vehicles Requiring a New PCM” section of this article for further details.

Obtaining the PCM Module Identification

1. With the New Generation Star (NGS) tester and the latest level Diagnostic Card (Red), use the following steps to obtain the PCM part number:
   a. Enter the Model Year and Engine Displacement of the vehicle.
   b. Select Diagnostic Data Link.
   c. Select Powertrain Control Module.
   d. Write down the PCM part number
2. Check if the PCM part number is listed in the chart found in Figure 1.
3. All 2000 and some 2001 model year vehicles will have a PCM included in the list. If the PCM part IS included in the list, refer to either “V6 Vehicles Requiring a New PCM” or “V8 Vehicles Requiring a New PCM” for the repair procedure.
4. Some 2001 model year vehicles will not be in the list. These vehicles only require that the PCM be reprogrammed to the latest level. Refer to the PCM Calibration Chart for details. No further service is required.

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V6 Vehicles Requiring a New PCM - Service Procedure

1. Disconnect the battery.
2. Replace the Powertrain Control Module (PCM) with the correct part which can be found in the chart in Figure 1. Refer to Workshop Manual, Section 303-14 for additional information.

NOTE
A PARAMETER RESET MUST BE PERFORMED FOR THE PASSIVE ANTI THEFT SYSTEM (PATS) TO OPERATE. FAILURE TO DO SO WILL RESULT IN A NO START CONDITION.

NOTE
DO NOT ERASE KEYS. IT MAY BE NECESSARY TO CYCLE THE KEY FROM OFF TO RUN 3 TIMES, EACH TIME LEAVING THE KEY IN THE RUN POSITION SO THE PATS MODULE WILL STORE THE NEW PCM ID.

3. Replace the Idle Air Control (IAC) Valve, Gasket and Bolts. IAC part 1F2Z-9F715-AA, gasket part E83Z-9F670-A, bolts part W704751-S309. Torque bolts to 89 Lb-in (10 N•m). Leave the electrical connector unplugged. Refer to Workshop Manual, Section 303-14 for additional information.

NOTE
DISCARD IAC SPACER (IF EQUIPPED).

4. Install the IAC Jumper Harness, part number 2W4Z-14A411-DA, by using the following procedure:
   a. Lift the Front Power Distribution Box (underhood) by disengaging 3 (three) retaining clips securing it to the vehicle (2 in rear - 1 in front). Gently lift and roll the box forward exposing the bottom cover.
   b. Remove the bottom cover of the box by releasing the 5 (five) retaining clips located along the sides of the cover.
   c. Position the box right side up and open the top cover. Remove the now exposed bolt securing the electrical connection to the rear/inside of the box. Refer to Figure 2.
   d. Route the eyelet end of the 2W4Z-14A411-DA Jumper Harness under the rear/inside section of the box and position the eyelet over the electrical connection.
   e. Install the retaining bolt. Torque to 105 Lb-in (12 N•m).
   f. Install the lower cover on the box. Carefully tuck all the wires behind the cover so they will not be pinched between the cover and the box.
   g. Carefully route the Jumper Harness around the front of the box. Avoid sharp edges and any opportunity to pinch or trap the harness. Refer to Figure 3.
   h. Install the box by sliding back over the brackets and engage the retaining clips.
   i. Route the Jumper Harness parallel with the right inner fender. Secure the first Christmas Tree fastener in the vacant hole in the inner fender. Refer to Figure 4.
   j. Remove the push pin fastener from far right side of the cowl leaf screen. Lift the right edge slightly and route the Jumper Harness into the engine compartment between the hood to cowl seal and the strut tower brace. Refer to Figure 5.
   k. Route the Jumper Harness along the existing engine harness, securing in 2 places with tie strap. Refer to Figure 6.
   l. Route the Jumper Harness under the intake manifold and behind the intake manifold support bracket located in the middle of the right cam cover. Secure the Jumper Harness to the EGR vacuum harness with a tie strap. Refer to Figure 6.
   m. Route the Jumper Harness along the right cam cover and tie strap to the small wiring harness which is secured to the lower front attaching stud on the cam cover. Refer to Figure 7.

NOTE
DO NOT ALLOW THE JUMPER HARNESS TO MAKE CONTACT WITH THE EGR TUBE OR DPFE HOSES.

   n. Pass the Jumper Harness under the upper radiator hose and connect to the IAC. Connect the original IAC connector/harness to the folded back connector on the Jumper Harness. Refer to Figure 7.

o. Connect the battery. Check for and clear any Diagnostic Trouble Codes that may be stored in the PCM.

V8 Vehicles Requiring a New PCM - Service Procedure

1. Disconnect the battery.
2. Replace the Powertrain Control Module (PCM) with the correct part which can be found in the chart in Figure 1. Refer to Workshop Manual, Section 303-14 for additional information.

NOTE
A PARAMETER RESET MUST BE PERFORMED FOR THE PASSIVE ANTI THEFT SYSTEM (PATS) TO OPERATE. FAILURE TO DO SO WILL RESULT IN A NO START CONDITION.

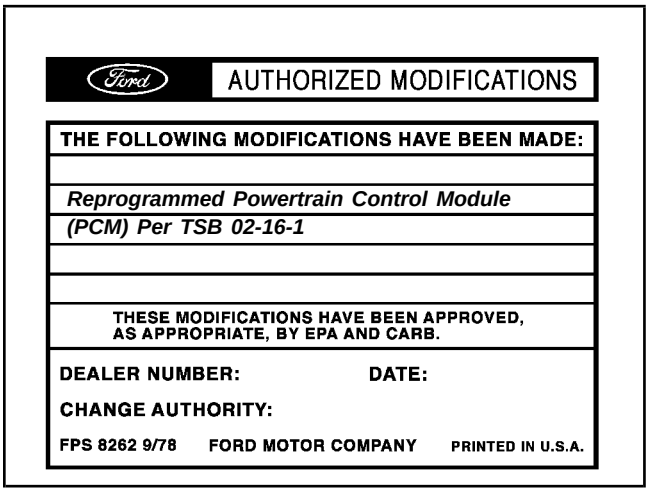
NOTE
DO NOT ERASE KEYS. IT MAY BE NECESSARY TO CYCLE THE KEY FROM OFF TO RUN 3 TIMES, EACH TIME LEAVING THE KEY IN THE RUN POSITION SO THE PATS MODULE WILL STORE THE NEW PCM ID.

3. Remove the engine appearance cover and the air induction tube. Refer to Workshop Manual, Section 303-12.
4. Install the IAC Jumper Harness, part number 2W4Z-14A411-CA, by using the following procedure:
   a. Lift the Front Power Distribution Box (underhood) by disengaging 3 (three) retaining clips securing it to the vehicle (2 in rear - 1 in front). Gently lift and roll the box forward exposing the bottom cover.
   b. Remove the bottom cover of the box by releasing the 5 (five) retaining clips located along the sides of the cover.
   c. Position the box right side up and open the top cover. Remove the now exposed nut securing the electrical connection to the rear/inside of the box. Refer to Figure 2.
   d. Route the eyelet end of the 2W4Z-14A411-CA Jumper Harness under the rear/inside section of the box and loop the eyelet over the electrical connection stud.
   e. Install the retaining nut. Torque to 105 Lb-in (12 N•m).
   f. Install the lower cover on the box. Carefully tuck all the wires behind the cover so they will not be pinched between the cover and the box.
   g. Carefully route the Jumper Harness around the front of the box. Avoid sharp edges and any opportunity to pinch or trap the harness. Refer to Figure 3.
   h. Install the box by sliding back over the brackets and engage the retaining clips.
   i. Route the Jumper Harness parallel with the right inner fender. Secure the first Christmas Tree fastener in the vacant hole in the inner fender. Refer to Figure 4.
   j. Remove the push pin fastener from far right side of the cowl leaf screen. Lift the right edge slightly and route the Jumper Harness into the engine compartment between the hood to cowl seal and the strut tower brace. Refer to Figure 5.
   k. Secure the second Christmas Tree fastener into the vacant hole under the far right side of the cowl leaf screen. Reposition the cowl leaf screen and install the push pin fastener. Refer to Figure 5.
   l. Secure the third Christmas Tree fastener into the vacant hole on top of the in-line electrical connector bracket located in the right rear of the engine compartment. Continue to route the Jumper Harness along the engine compartment wiring harness. Secure with a strip tie. Refer to Figure 8.
   m. Secure the fourth Christmas Tree fastener to the vacant hole in the bracket attached to the right fuel rail. Refer to Figure 9.
   n. Connect the Jumper Harness to the IAC and the original IAC connector/harness to the folded back connector on the Jumper Harness. Refer to Figure 9.
   o. Install the air induction tube and the engine appearance cover.
   p. Connect the battery. Check for and clear any Diagnostic Trouble Codes that may be stored in the PCM.

PCM CALIBRATION CHART							
Applications	Old Part Number (-12A650-)	Old Tear Tag	New Part Number (-12A650-)	New Tear Tag	Old Calibration	New Calibration	NGS/WDS Qualifier
2002 LS V6 MAN TRANS SPORT	2W4A-ME	YC04	2W4Z-MF	YC05	2LQ26S0511	2LQ26S0512	WDS B19A.10 Release or Later
2001 LS V6 MAN TRANS SPORT	1W4A-ZA	UVK0	1U7Z-EVA	VABO	1LQ26S0512	1LQ26S0512	WDS B19A.10 Release or Later
2002 LS V6 AUTO BASE	2W4A-ND	LK03	2W4Z-NE	LK04	2LQ16B0511	2LQ16B0512	WDS B19A.10 Release or Later
2002 LS V6 AUTO BASE	2W4A-VC	CCG2	2U7Z-BZA	ZTB0	2LQ17B0510	2LQ17B0510	WDS B19A.10 Release or Later
2001 LS V6 AUTO BASE	1W4A-BAA	BZI0	1U7Z-EXA	XCG0	1LQ16B0512	1LQ16B0512	WDS B19A.10 Release or Later
2002 LS V6 AUTO SPORT	2W4A-PD	KGV3	2W4Z-PE	KGV4	2LQ16S0511	2LQ16S0512	WDS B19A.10 Release or Later
2002 LS V6 AUTO SPORT	2W4A-XC	UAH2	2U7Z-CAA	ADH0	2LQ17S0510	2LQ17S0510	WDS B19A.10 Release or Later
2001 LS V6 AUTO SPORT	1W4A-CAA	LTG0	1U7Z-EYA	YFC0	1LQ16S0512	1LQ16S0512	WDS B19A.10 Release or Later
2002 LS V8 AUTO BASE	2W4A-BD	STH3	2W4Z-BE	STH4	2LQ18B0511	2LQ18B0512	WDS B19A.10 Release or Later
2002 LS V8 AUTO BASE	2W4A-CD	KUL3	2W4Z-CE	KUL4	2LQ18B0M11	2LQ18B0M12	WDS B19A.10 Release or Later
2002 T-BIRD BASE	2W6A-EC	HDE2	2W6Z-ED	HDE3	2SR12M0512	2SR12M0513	WDS B19A.10 Release or Later
2002 LS V8 AUTO SPORT	2W4A-DD	SEC3	2W4Z-DE	SEC4	2LQ18S0511	2LQ18S0512	WDS B19A.10 Release or Later
2002 LS V8 AUTO SPORT	2W4A-ED	EGT3	2W4Z-EE	EGT4	2LQ18S0M11	2LQ18S0M12	WDS B19A.10 Release or Later
2002 LS V8 AUTO BASE	2W4A-RB	PKE1	2U7Z-BRA	RMJ0	2LQ19B0510	2LQ19B0510	WDS B19A.10 Release or Later
2002 LS V8 AUTO BASE	2W4A-TB	RVG1	2U7Z-BSA	XAV0	2LQ19B0M10	2LQ19B0M10	WDS B19A.10 Release or Later
2002 T-BIRD V8 BASE	2W6A-DB	KNS1	2U7Z-BTA	TAJ0	2SR11M0510	2SR11M0510	WDS B19A.10 Release or Later
2002 LS V8 AUTO SPORT	2W4A-SB	PFJ1	2U7Z-BXA	BHB0	2LQ19SO510	2LQ19SO510	WDS B19A.10 Release or

Applications	Old Part Number (-12A650-)	Old Tear Tag	New Part Number (-12A650-)	New Tear Tag	Old Calibration	New Calibration	NGS/WDS Qualifier
2002 LS V8 AUTO SPORT	2W4A-UB	EYL1	2U7Z-BYA	CHD0	2LQ19S0M10	2LQ19S0M10	Later WDS B19A.10 Release or Later
2001 LS V8 AUTO BASE	1W4A-AMA	TPH0	1U7Z-EKA	ZHU0	1LQ18B0511	1LQ18B0511	WDS B19A.10 Release or Later
2001 LS V8 AUTO BASE	1W4A-APA	NEX0	1U7Z-ELA	DNC0	1LQ19B0511	1LQ19B0511	WDS B19A.10 Release or Later
2001 LS V8 AUTO SPORT	1W4A-ANA	FZG0	1U7Z-EHA	EMB0	1LQ18S0511	1LQ18S0511	WDS B19A.10 Release or Later
2001 LS V8 SPORT	1W4A-ARA	IBX0	1U7Z-EMA	FAM0	1LQ19S0511	1LQ19S0511	WDS B19A.10 Release or Later

Obtain an Authorized Modifications Decal (STAT 5941-2 - obtainable through the PDC) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.



PART NUMBER	PART NAME
1F2Z-9F715-AA	Idle Air Control (IAC) Valve
E83Z-9F670-A	IAC Gasket
W704751-S309	Bolt
2W4Z-14A411-DA	IAC Jumper - V6
2W4Z-14A411-CA	IAC Jumper - V8
1U7Z-12A650-ENA	PCM
1U7Z-12A650-EPA	PCM
XU7Z-12A650-ATA	PCM
1U7Z-12A650-ESA	PCM
XU7Z-12A650-AXA	PCM
1U7Z-12A650-EHA	PCM
1U7Z-12A650-EKA	PCM
1U7Z-12A650-ELA	PCM
1U7Z-12A650-EMA	PCM

OTHER APPLICABLE ARTICLES: NONE

WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
021601A	Reprogram Powertrain Control Module 2002 MY	0.4 Hr.
021601B	Obtain Powertrain Control Module Identification Then Reprogram PCM 2001 MY	0.4 Hr.
021601C	Obtain Powertrain Control Module Identification Then Replace Powertrain Control Module, And Idle Air Control Valve And Install IAC Jumper Harness 2001 MY 3.0L	1.6 Hrs.

Article No. 02-16-1 Cont'd.

021601D	Obtain Powertrain Control Module Identification Then Replace Powertrain Control Module, And Install IAC Jumper Harness 2001 MY 3.9L	1.7 Hrs.
021601E	Replace Powertrain Control Module, And Idle Air Control Valve Then Install IAC Jumper Harness 2000 MY 3.0L	1.6 Hrs.
021601F	Replace Powertrain Control Module, And Install IAC Jumper Harness 2000 MY 3.9L	1.6 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
12A650	42

OASIS CODES: 602300, 603300

3.9L V8 ENGINE PCM CHART

NEW PCM PART NUMBER	REPLACE OLD PCM
1U7Z-EHA	1U7A-FA 1U7A-FB 1U7A-WA 1W4A-ABA 1W4A-HA 1W4A-HB
1U7Z-EKA	1U7A-DA 1U7A-DB 1U7A-VA 1W4A-AAA 1W4A-GA 1W4A-GB
1U7Z-ELA	1U7A-GA 1U7A-GB 1U7A-YA 1W4A-ACA 1W4A-JA 1W4A-JB
1U7Z-EMA	1U7A-HA 1U7A-HB 1U7A-ZA 1W4A-ADA 1W4A-KA 1W4A-KB

3.0L V6 ENGINE PCM CHART

NEW PCM PART NUMBER	REPLACE OLD PCM
XU7Z-AXA	XU7A-SA
XU7Z-ATA	XU7A-TA
1U7Z-ENA	1U7A-NB 1U7A-SA 1W4A-AGA 1W4A-AGB 1W4A-EB
1U7Z-EPA	1U7A-JA 1U7A-JB 1U7A-TA 1W4A-AC 1W4A-AEA 1W4A-AEB
1U7Z-ESA	1U7A-KA 1U7A-KB 1W4A-AFA 1W4A-AFB 1W4A-BC 1U7A-UA

TB-6677-A

Figure 1 - Article 02-16-1

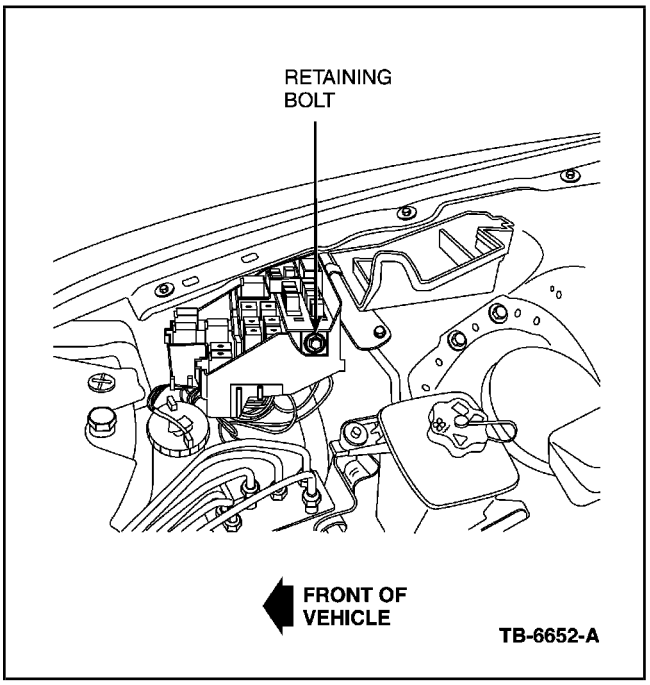


Figure 2 - Article 02-16-1

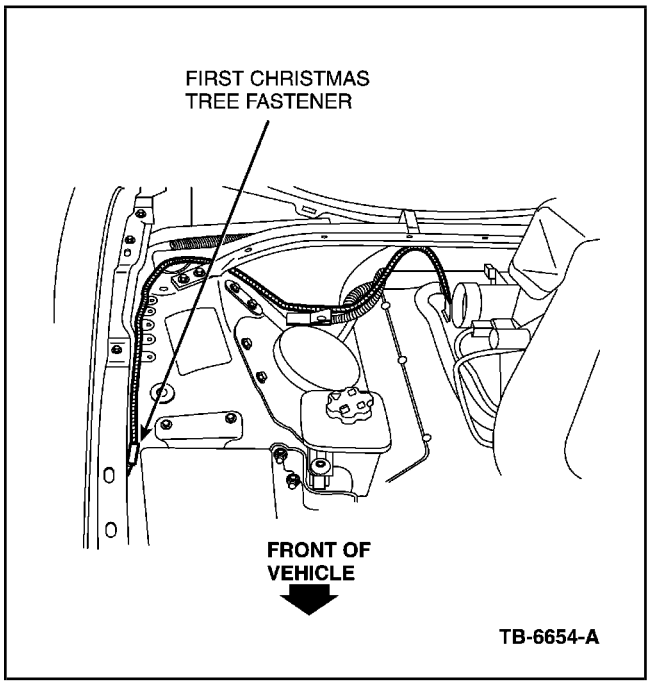


Figure 4 - Article 02-16-1

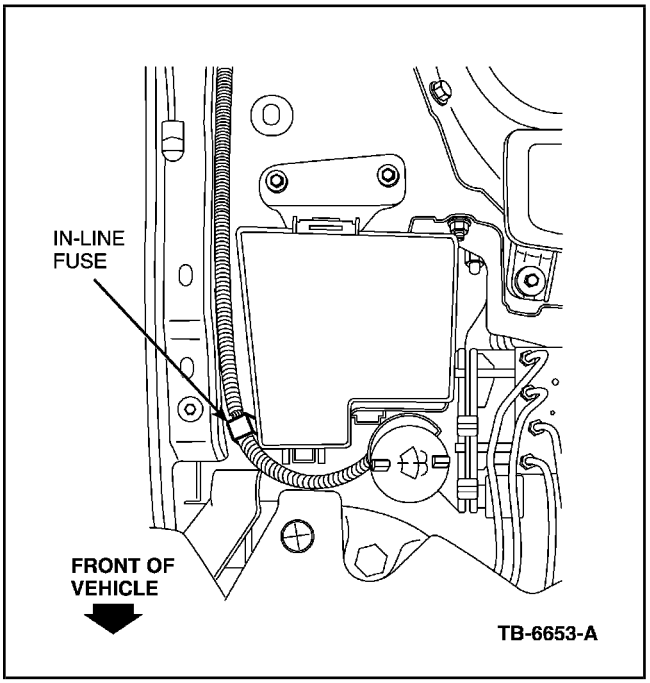


Figure 3 - Article 02-16-1

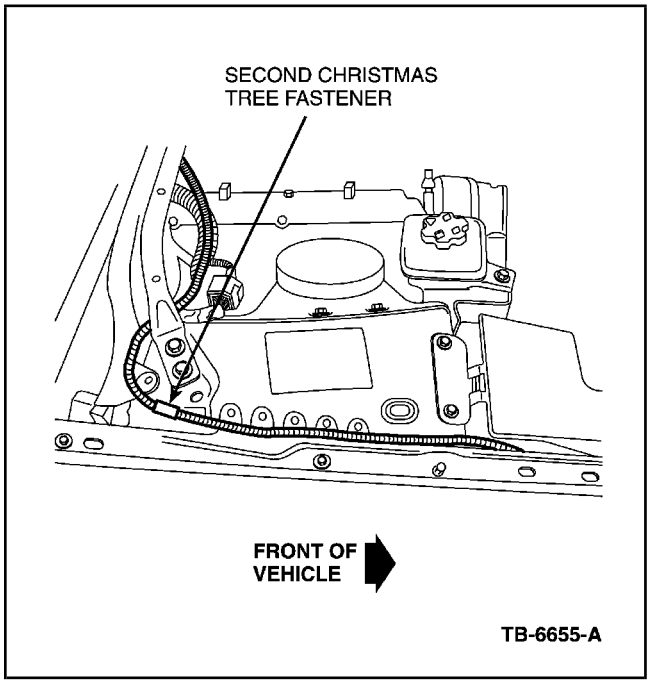


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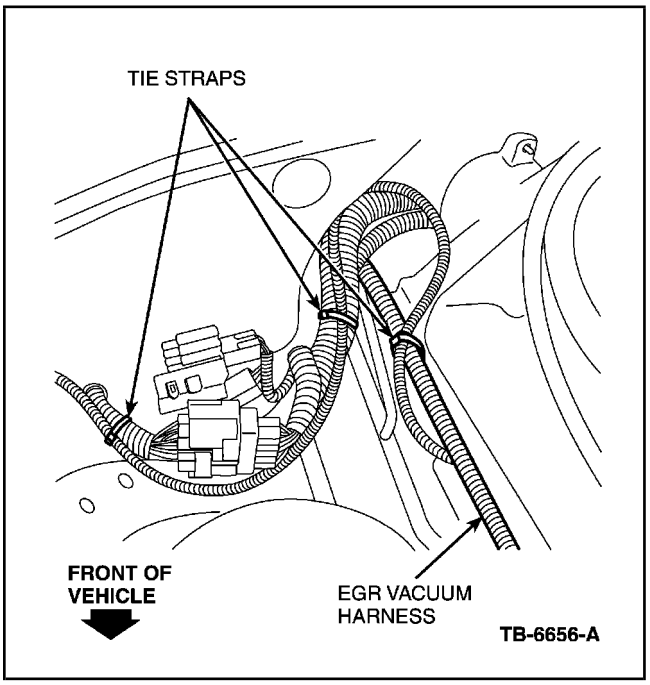


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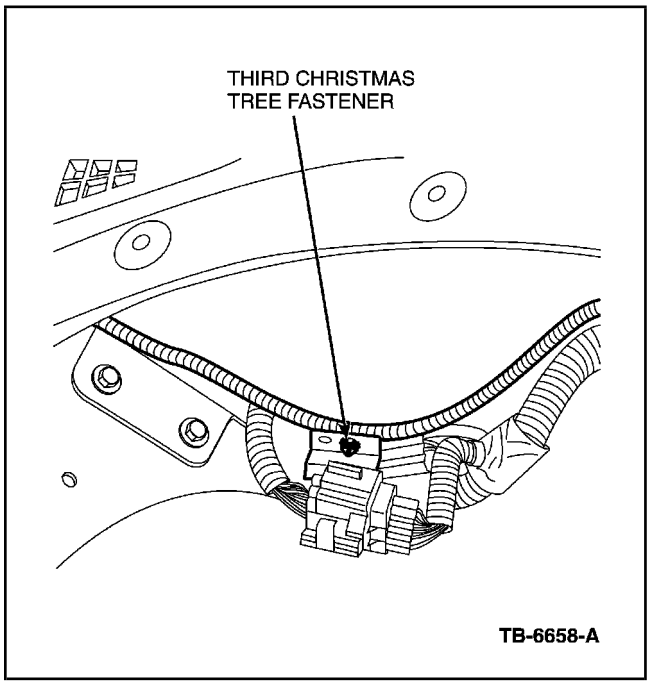


Figure 8 - Article 02-16-1

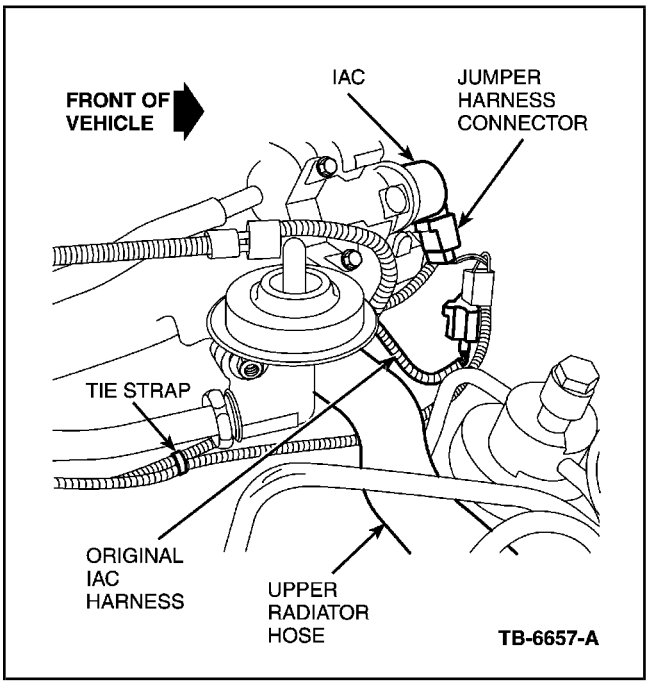


Figure 7 - Article 02-16-1

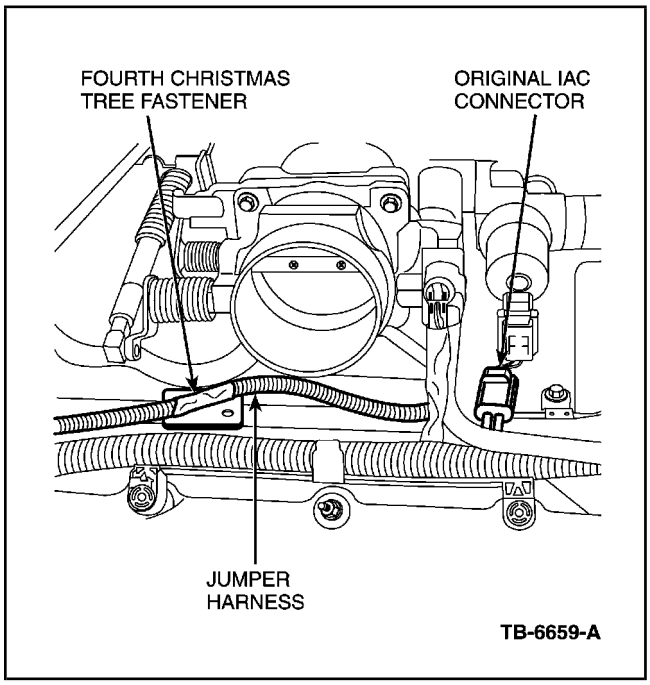


Figure 9 - Article 02-16-1